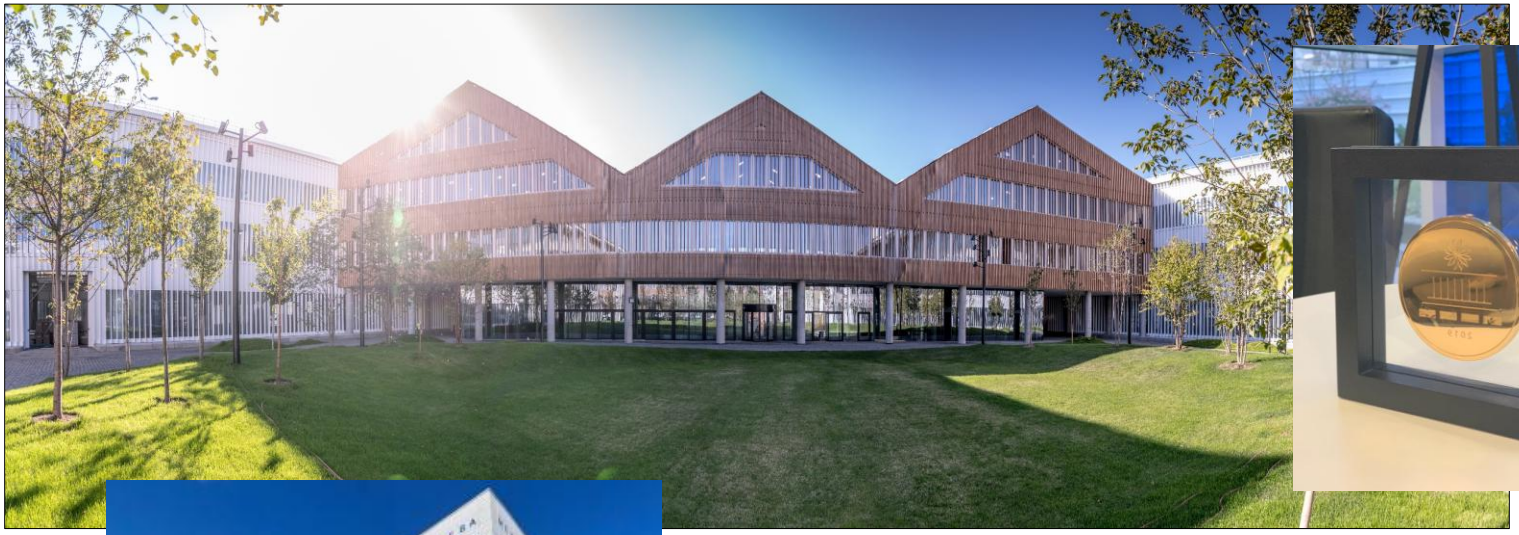
Materials Science
M.Sc. Program



Prof. Keith J. Stevenson

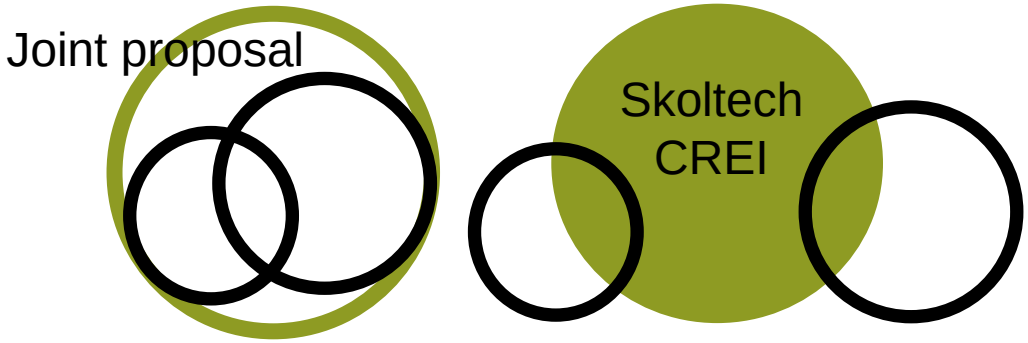
April 2, 2019

Open Doors

Skoltech at a Glance

- №1 in Research in the Russian Federation
- English language
- 2019: 140+ faculty
  1100+ students (MSc and PhD)
- 10 MSc and 7 PhD programs accredited in the Russian Federation (3 internationally by HCÉRES)
- Listed in Top-100 Nature Index 2019 Young University ranking
- World-class research infrastructure in target domains
- 40 PhD Thesis defenses held

Centers of Research, Education and Innovation (CREI)



Pursue Innovative Research

Deliver Scientific Growth

Generate World-Class Educational Programs

CEE: Electrochemical Energy Storage

est. 2013

Electrochemical energy storage technologies & materials

Computational materials science

Solar energy conversion

CEST: Energy Science and Technology

formed 2018 as a merger with Center of Energy Systems

Power electronics, smart grids, integrated energy infrastructures, energy markets and regulation

Centers of Research, Education and Innovation (CREI)

Data Science & Artificial Intelligence	**CDISE**: Computational and Data-Intensive Science
Life Science & Health	**CLS**: Life Sciences **CNBR**: Neurobiology and Brain Restoration
Cutting-Edge Engineering & Advanced Materials	**CDMM**: Design, Manufacturing and Materials **CHR**: Hydrocarbon Recovery **CS**: Space
Energy Efficiency	**CEST**: Energy Science and Technology
Photonics & Quantum Technologies	**CPQM**: Photonics and Quantum Materials
Advanced Studies	**CAS**: Advanced studies

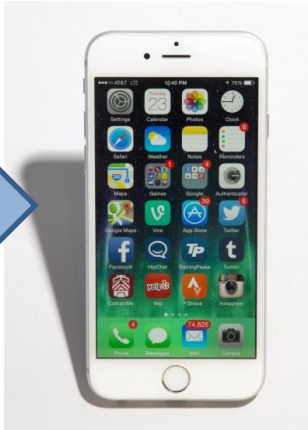
Research-oriented

M.Sc. Program at a Glance

- 16 faculty supervisors from 4 CREIs
- 30 students (20% international)
- 2019: >10 candidates per place at admission
- >30 M.Sc. defenses held
- > 1000 research papers co-authored, >10 patents filed
- 12 recommended courses
- access to 8 operational research laboratories and to shared research facilities
- networks of academic and industrial partners
- external grants and contracts
- https://crei.skoltech.ru/cest/education/materials-science-program/

What is (modern) Materials Science?

Would not exist without recent advances in Materials Science

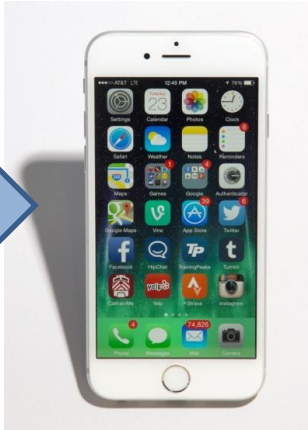


Elementary devices/parts

- box/frame
- processor (transistors)
- ???

What is (modern) Materials Science?

Would not exist without recent advances in Materials Science



Elementary devices/parts

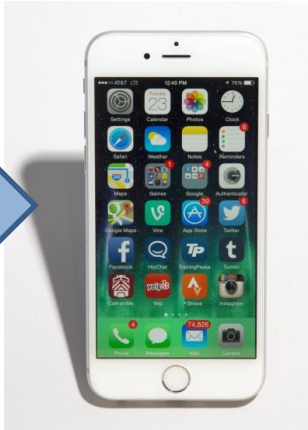
- box/frame
- processor (transistors)
- display
- battery
- thermal management
- wires, antennas, audio
- …

Functional materials

- mechanical & chemical
- semiconductors
- ???

What is (modern) Materials Science?

Would not exist without recent advances in Materials Science



Elementary devices/parts

- box/frame
- processor (transistors)
- display
- battery
- thermal management
- wires, antennas, audio
- …

Functional materials

- mechanical & chemical
- semiconductors
- light emitter
- battery electrodes
- thermal conductors
- thermal insulators
- …

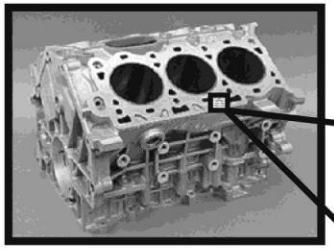
What do Materials Scientists do?

OLED vs LCD visualization

Above: Simulation of the difference in contrast from the AC7 OLED to a traditional LCD.

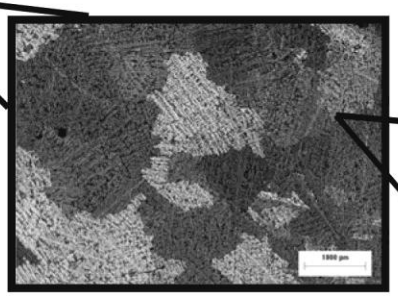
- Create new materials
- Optimize existing materials
- Create/study/optimize elementary devices (lab prototypes)

Materials Science vs. Materials Engineering



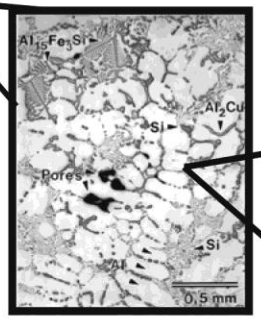
Macro-Scale Structure
Engine Block
≅ upto 1 meter

Performance Criteria
- **Power generated**
- **Efficiency**
- **Durability**
- **Cost**



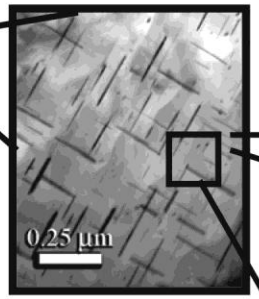
Microstructure
- Grains
≅ 1 – 10 millimeters

Properties affected
- High cycle fatigue
- Ductility



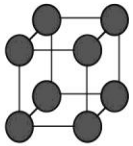
Microstructure
- Dendrites & Phases
≅ 50 – 500 micrometers

Properties affected
- Yield strength
- Ultimate tensile strength
- High cycle fatigue
- Low cycle fatigue
- Thermal Growth
- Ductility

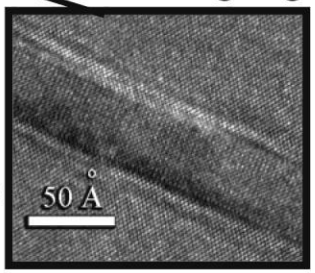


Nano-structure
- Precipitates
≅ 3-100 nanometers

Properties affected
- Yield strength
- Ultimate tensile strength
- Low cycle fatigue
- Ductility



Unit Cell



Atomic-scale structure
≅ 1-100 Angstroms

Properties affected
- Young's modulus
- Thermal Growth

Materials Science

Materials Engineering

A real-world example of important microstructural features at different length-scales, resulting from the sophisticated synthesis and processing used, and the properties they influence. The atomic, nano, micro, and macro-scale structures of cast aluminum alloys (for engine blocks) in relation to the properties affected and performance are shown. The materials science and engineering (MSE) tetrahedron that represents this approach is shown in the upper right corner.

(Illustrations Courtesy of John Allison and William Donlon, Ford Motor Company)

Materials Science Track: Courses

Core courses

- Survey of Materials **(Term 1B)**
- Materials Chemistry **(Term 2)**

Experimental Materials Science

Recommended courses

- Introduction to Solid State Physics**(2)**
- Material Structure Characterization Methods **(3)**
- Organic Materials for Electronics, Energy Conversion and Storage **(3)**
- Carbon Nanomaterials **(3)**
- Electrochemistry **(4)**

Computational Materials Science

Recommended courses

- Computational Chemistry and Materials Modeling **(2)**
- Structure and Property of Materials **(3)**
- Advanced Materials Modeling **(4)**
- Advanced Solid State Physics **(6)**
- Materials Selection in Design **(5)**

Elective courses

- discuss with your advisor

Faculty

Artem Abakumov
Full Professor, Director

Alexander Shapeev
Associate Professor

Sergey Abaimov
Assistant Professor

Igor Shishkovsky
Associate Professor

CDMM: Nanocomposite Materials & Thermal Barrier Coatings

Evgeny Antipov
Full Professor

Keith Stevenson
Full Professor, Provost

Albert Nasibulin
Full Professor

CPQM: Synthesis and Applications of Nanomaterials

Alexei Buchachenko
Full Professor

Pavel Troshin
Associate Professor

Sergey V. Levchenko
Assistant Professor

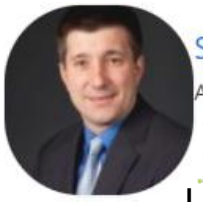
Andriy Zhugayevych
Assistant Professor

Timofei Zatsepin
Associate Professor

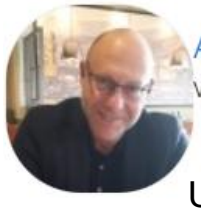
CLS: Electrochemical Biosensors

Artem R. Oganov
Full Professor

Adjunct & Visiting Faculty, Researchers, Staff



Sergei Tretiak
Adjunct Professor
✓✓
LANL



Alexander Korsunsky
Visiting Professor
University of Oxford

Xavier Gonze
Visiting Professor
UC Louvain

Gleb Sukhorukov
Visiting Professor
Queen Mary U London

Victoria Nikitina
Research Scientist

Stanislav Fedotov
Senior Research Scientist

Anna Sharova
Junior Research Scientist
✓

Polina Rumyantseva
Research Program Manager

Dmitry Aksenov
Senior Research Scientist

Alexander Kvashnin
Senior Research Scientist

Dmitry Krasnikov
Research Scientist
CPQM

Fedor Fedorov
Senior Research Scientist
CPQM

Stanislav Evlashin
Senior Research Scientist
CDMM

Lyubov Frolova
Senior Research Scientist

Oleg Drozhzhin
Senior Research Scientist

Svetlana Lipovskikh (Borisova)
Research Scientist

Yury Gladush
Senior Research Scientist
CPQM

Artem Zaikin
Laboratory Engineer

Dmitry Pogozhev
Industry Project Manager

Victor Vysotsky, Alexei Salimon … and many others!

M.Sc. Research in Materials Science

In brief

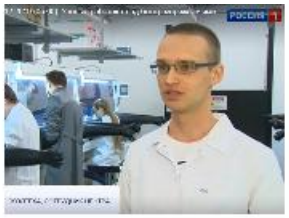
- *Highly interdisciplinary*
- Materials for energy storage – Abakumov, Antipov, Fedotov, Nikitina, Stevenson
- Optoelectronic materials – Nasibulin, Troshin
- Structural materials – Korsunsky
- Computational Materials Science – Gonze, Levchenko, Oganov, Shapeev, Tretiak, Zhugayevych
- Theoretical Materials Science – Buchachenko, Fine
- … *with details on faculty web-pages: example*

M.Sc. Thesis 2016-2019

- Materials for energy conversion and storage (19 students)
- Materials for optoelectronics (8 students)
- Computational materials science (4 students)
- Other materials (2 students)

At Skoltech you will do world-level research at cutting edge of science

Broad range of topics: see e.g. public media reports



4/12/2019

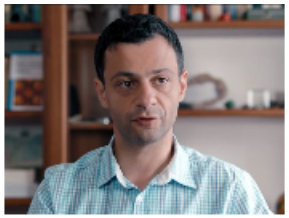
Skoltech PhD student, Roman Kapaev, on potassium batteries. Watch on Rossiya 1

28/11/2019

Albert Nasibulin and his colleagues have created a transparent film based on nanotubes. Read on Izvestia, popmech.ru

11/11/2019

A group of scientists led by Pavel Troshin have developed superfast charging high-capacity potassium batteries based on organic. Read on EurekAlert!, Indicator, Poisknews, popmech, phys.org, newinform, openscience.news, bitcryptonews, pv-magazine, azom.com



7/11/2019

A group of scientists led by Artem Oganov have succeeded in synthesizing thorium decahydride (ThH10), a new superconducting material with a very high critical temperature (161 K). Read on TASS, Naked Science, Interfax, Indicator, Regnum, N+1, Yandex.news, news.sputnik.ru, sib-science.info, rscf.ru, scitechdaily.com, techsite.io, nanotech-now.com, Phys.org, chemeurope.com

31/10/2019

Skoltech Ph.D. student, Olga Kraevaya and her colleagues from the Institute for Problems of Chemical Physics of Russian Academy of Sciences and National Taiwan University have discovered that fullerene compounds can help in the fight against lung cancer. Read on TASS, EurekAlert!, NakedScience, popmech.ru, meddaily.ru

23/08/2019

Alexei Salimon and his colleagues from IPAC RAS and MISIS developed a new technology for producing ultra-high molecular weight polyethylene with a porous structure. Read on TASS, Rambler.news, Indicator, Gazeta.ru, mk.ru, news.Sputnik, nauka.vesti.ru

7/08/2019

Alexander Kvashnin and his colleagues presented the research optimizing the search for new materials that can replace diamonds in drilling rigs. Read on Phys.org, Nauchnaya Rossiaya, Vechernyaya Moskva, RIANovosti, Chrdk., goroday.ru, bitcryptonews.ru

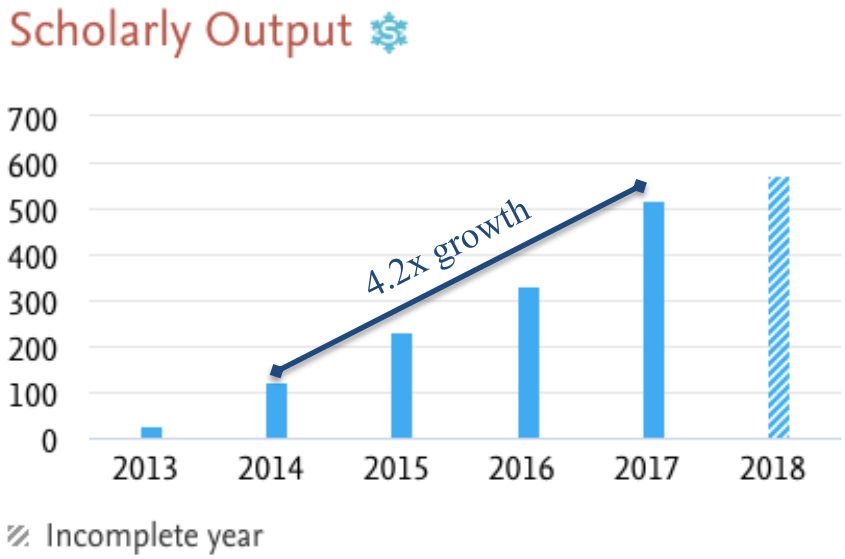
3/06/2019

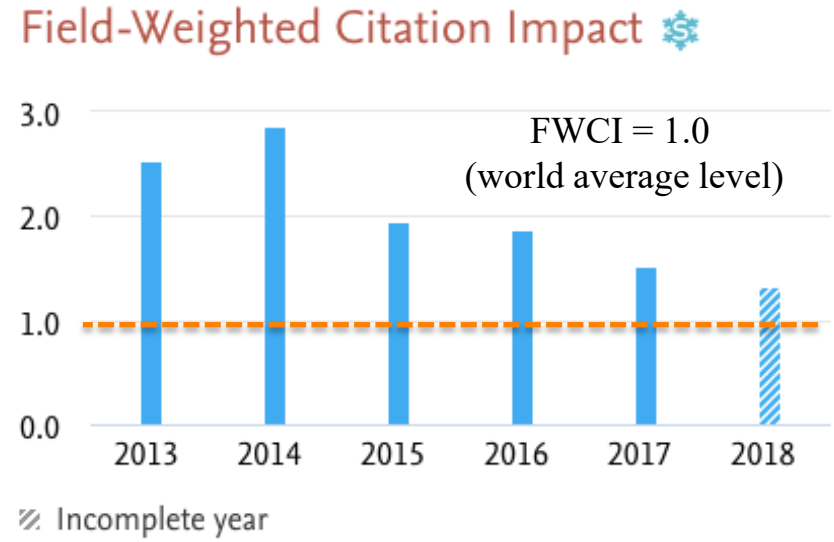
Skoltech scientists have created a new polymer cathode material for fast-charged metal-ion batteries that surpasses all previous similar developments. Read on Naked Science, RIANovosti, Indicator

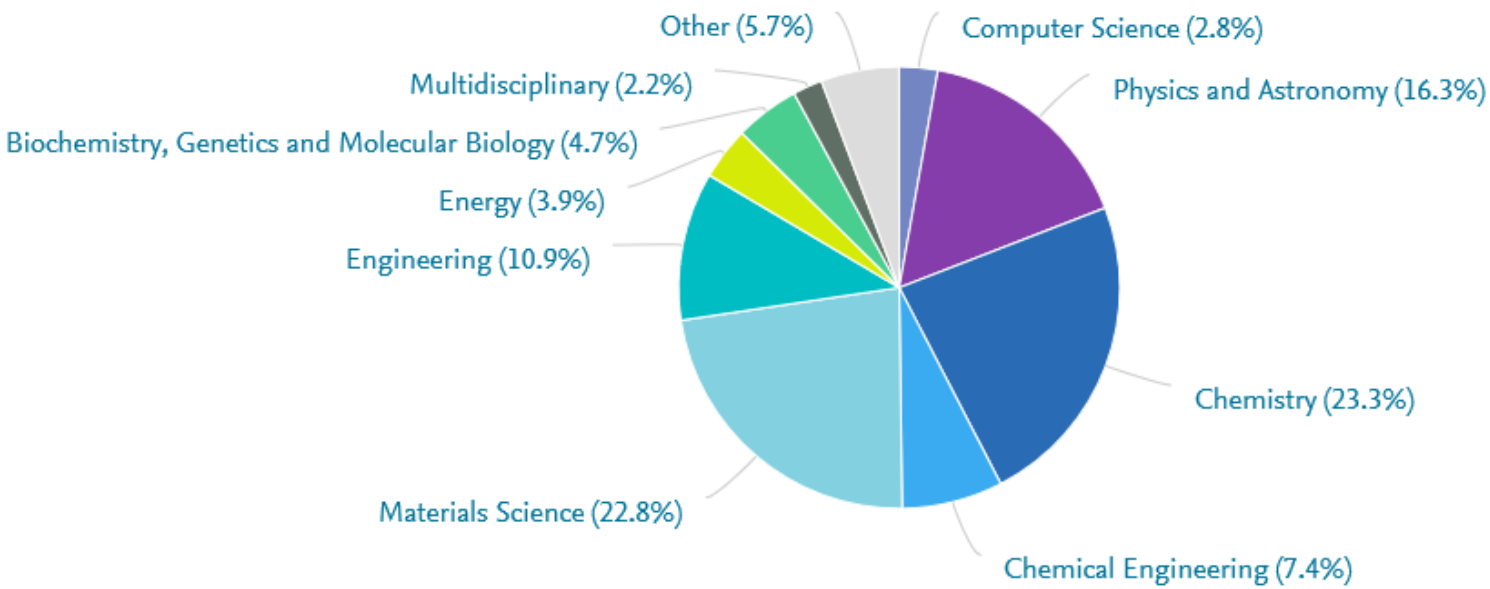
26/05/2019

Skoltech scientists and their American colleagues have created artificial intelligence, accelerating quantum calculations. Read on RIANovosti, scientificrussia.ru

Research Prominence in Material Science







On-site Research Facilities

CEST Labs

Cell assembling/
Structure characterization

Material processing lab

Electrochemistry Analytical lab

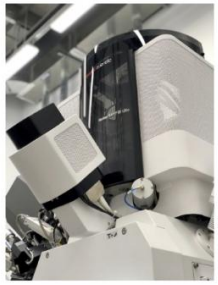
CPQM

Laboratory of NanoMaterials

Shared facilities

ADVANCED IMAGING CORE FACILITY

AICF



FABLAB AND MACHINE SHOP

FAB LAB

Organization & Management

Program	**Program Committee**, EDU committee	**Education Department**
Admission	**Program Committee,** Supervisor, EDU committee	**Student Outreach & Recruitment**
Student's trajectory	**Student**, **Supervisor**, IDC	**MSc and Doctoral Study Office**
Courses	**Student**, **Course Instructors**, Program Committee (Qualif. Exam), students	**Education Department**, **Disciplinary Board**
Research	**Student**, **Supervisor**, IDC, Program Committee	**MSc and Doctoral Study Office**
Research outcomes	**Student**, **Supervisor,** IDC peer review (publ. & conf.) Program Committee+External Chair (FTR) **MSc and PhD Defense**	**MSc and Doctoral Study Office** + other administrative streams
	Student, **Supervisor,** IDC **Program Committee**	**MSc and Doctoral Study Office** **Disciplinary Board**

Funding

Tuition fee		Appointment, Promotion & Tenure Committee
Stipend		EDU Committee/Dean of EDU
Research	Infrastructure	Research & Innovation Committee/CREI
		Partners
	Consumables	External funding: Grants & Contracts
		CREI/Faculty startup funds
		Competitive Skoltech programs /NGP, STRIP SBI
	Academic mobility	External funding: Grants & Contracts
	Additional payments	External funding: Grants & Contracts

External Funding

RUSSIAN FOUNDATION FOR BASIC RESEARCH

Russian Science Foundation

VolkswagenStiftung

Energynet
Национальная технологическая инициатива

ГАЗПРОМ НЕФТЬ	Novel super hard materials; **2019**
ГАЗПРОМ СТРОЙТЭК САЛАВАТ	Development of testing technology for soil heat stabilizers; **2019**
HUAWEI	Thermal conductive materials; **2019**
LG Chem	Materials for lithium ion batteries; **2017-2019**
QuintTech	Materials for solar panels; **2017**
РОСАТОМ	Radiation damage **2019 pending**

Integration into Job Market

Education-related activities

- 2-month Industrial Immersion (MSc)
- 2-month Academic Mobility
- Industry Day

Research-related activities

- Research grants and Industrial contracts
- Scientific partners and collaboratiors

Entrepreneurship-related activities

- Startups (E&I Center, Skolkovo foundation)

Local events

- Skolkovo Innovation Center events
  - Startup Village, Open Innovations, Job and the City etc.
- Skoltech conferences (meet Industry and Academic representatives)
  - International School-Conference of Young Scientists (energy storage materials)
  - International School-Conference in Computational Materials Science
  - other Skoltech conferences in Materials Science and Engineering

Academic Partners

Massachusetts Institute of Technology

IPCP RAS

Universiteit Antwerpen

MIPT MOSCOW INSTITUTE OF PHYSICS AND TECHNOLOGY

Aalto University

COLLÈGE DE FRANCE 1530

THE UNIVERSITY OF TEXAS AT AUSTIN

NORTHWESTERN POLYTECHNICAL UNIVERSITY

Novosibirsk State University *THE REAL SCIENCE

UNIVERSITAS CATHOLICA LOVANIENSIS SEDES SAPIENTIAE 1425

Newcastle University

ОБЪЕДИНЕННЫЙ ИБТРАН

UNIVERSITY AT ALBANY State University of New York

TALLINN UNIVERSITY OF TECHNOLOGY

THE UNIVERSITY OF TEXAS AT DALLAS EST. 1969

TECHNISCHE UNIVERSITÄT MÜNCHEN

Partnership Networks

University	Country	Field-weighted citation impact	Outputs in Top 25 citation percentile (%)	Collaboration (%)	Collaboration impact (%)	Field-weighted views impact	Outputs in Top 25 views percentile most viewed (%)
Skolkovo Institute of Science and Technology	Russia	1.42	47.2	**68.5**	**7.2**	**2.29**	**73.2**
Massachusetts Institute of Technology	USA	2.12	54	55.7	13	1.55	52
Stanford University	USA	2.52	53.5	52.1	14.8	1.35	47.3
University of California at Berkeley	USA	2.33	54.9	53.4	13.3	1.45	48.9
Georgia Institute of Technology	USA	2.17	52.9	52.2	16	1.42	51.5
University of California at Los Angeles	USA	2.09	51.3	53.8	13.2	1.31	47
University of Texas at Austin	USA	2.03	49.8	43.3	13.8	1.32	46.6
California Institute of Technology	USA	1.97	44	40.6	9.3	1	31
King Abdullah University of Science and Technology	SA	2.35	61.8	**68.1**	**14.4**	**2.75**	**74.1**
University of Cambridge	UK	1.9	52.9	67.6	11.8	1.57	56.2
Hong Kong University of Science and Technology	China	2.22	57.8	73.4	14	1.69	67.4
Nanyang Technological University	China	2.18	53.8	66	13.7	1.39	54.3
Technical University of Munich	Germany	1.6	44.1	54.2	9.1	1.41	49.3
Ecole Polytechnique	France	1.16	39.4	60.8	7.7	1.13	40.1
Lomonosov Moscow State University	Russia	0.75	27.7	37.6	5.8	1.65	50.8
St. Petersburg State University	Russia	0.76	29.8	41	5.9	2.07	66.5
5-100 (Russia)	Russia	0.85	23.1	31.8	5	2.17	62
ITMO St. Petersburg	Russia	1.02	27.2	49.1	4.8	**2.31**	**57.9**
Tomsk Polytechnic University	Russia	0.96	17.9	21.4	4.4	2.23	65.2
Novosibirsk State University	Russia	0.75	31.4	33.8	6	2.23	61.9

Impact and output metrics of the selected schools and "startup Universities" in the field of *Materials Science* (Elsiever SciVal, 2015-2019)

Industrial Partners

Startup company “K-plus” Ltd*

The goal of the project is a commercialization of promising electrochemical systems based on K-ion batteries. Develop technologies for cathode materials based on $K_xM[B(CN)_6]$ (where M, B are Fe, Co, Cr, Mn) cyanide transition metal complexes, and high-capacity carbon-based anode materials.

Startup company “Rustor” Ltd**

The main areas of work are cathode materials based on phosphates with the structure of olivine ($LiFePO_4$, $LiFe_{1-x}Mn_xPO_4$, also known as LFP and LFMP) and layered oxides ($LiNi_{1-x-y}Mn_xCo_yO_2$, also known as NMC).

Startup company “Core Technology”

Developed code USPEX allows to predict material crystal structure with arbitrary P-T conditions by knowing only chemical composition of the material. Company provides services to commercial companies in a field of prediction of structure and chemical composition of functionalized materials for different application.

In stage of application for SKOLKOVO residence* * Resident of SKOLKOVO*

Career paths

Training time

- (PhD+) **Research & teaching:**
  - Academic universities (e.g. Skoltech)
- (PhD) **Research:**
  - Universities (Skoltech, MIT)
  - Research institutions (Los Alamos Nat Lab)
  - R&D labs in large corporations (Samsung)
  - other high-tech companies
- (MSc) **Engineering, manufacturing, support etc.:**
  - Startups (batteries, solar cells, electronics)
  - Industry (energy, electronics, chemical)

Skoltech is the right place to start your career of a Materials Scientist or Engineer

Job market: Examples from high-tech industry

Advanced materials

- Boeing, Mitsubishi
- IMEC

Energy storage

- Tesla Motors
- Liotech

Energy conversion

- Heliatek, Solarmer

Electronics

- Samsung, IBM
- Plastic Logic

Chemical

- BASF, Chevron, DuPont

Automotive

- Nissan, KAMAZ

IT (materials modeling)

- Continuum Analytics, Gaussian, MedeA, Simbeyond
- Kintech (Skolkovo resident)

Skoltech will make you competitive on this market

Job market: Established partners for CEST CREI

- Multiple joint projects with MIT, MSU, MIPT

Joint Labs

- with MSU “Advanced Materials for Green Energy”
- with Institute of Problems of Chemical Physics
- with Mendeleev University

Academic – research and internships

- Los Alamos National Lab, University of Texas at Austin, Nanjing University, Johannes Kepler University

Academic – research

- University of Bayreuth (Volkswagen), University of Antwerp (research contract), University of Ulm (Helmholtz Institute), University at Albany (CRDF)
- EU framework project COST Action 1307 StableNextSol

Industry

- InEnergy, Systems for Microscopy and Analysis, Liotech, TAGRAS, KAMAZ
- LG, Bosch, Nissan, Samsung

Small companies

- SuperOx, Skolkovo startups

At Skoltech you will have access to partners via Research & Industrial “immersion”

PhD Graduates That First Started as MSc

Dr. Andrey Chekannikov (Prof. Keith Stevenson)
Na-ion battery electrode materials based on the polyanionic compounds (Cand.Sci., IPCh&ECh, 2018)
MISiS

Skoltech
Skolkovo Institute of Science and Technology

Dr. Evgenia Gilshteyn (Prof. Albert Nasibulin, CPQM)
Components for Stretchable Electronics based on Single-Walled Carbon Nanotubes (Skoltech, 2018)
Empa - Swiss Federal Laboratories for Materials Science & Technology

Empa
Materials Science and Technology

Dr. Aliya Glagoleva (Prof. Keith Stevenson, Dr. Vasily Borzenko, JIHT RAS)
Development of kW Scale Hydrogen Energy Storage System (Skoltech, 2018)
Skoltech

Skoltech
Skolkovo Institute of Science and Technology

Dr. Konstantin Gubaev (Prof. Alexander Shapeev)
Machine-learning interatomic potentials for multicomponent alloys (Skoltech, 2019)
TU Delft

TU Delft
Delft University of Technology

Welcome to Skoltech

Further information:

- ➜ Lab Tour
- ➜ crei.skoltech.ru/cest/education/materials-science-program
- ➜ msc.skoltech.ru/materials-science
- ➜ If you are interested to join a specific research project contact our faculty